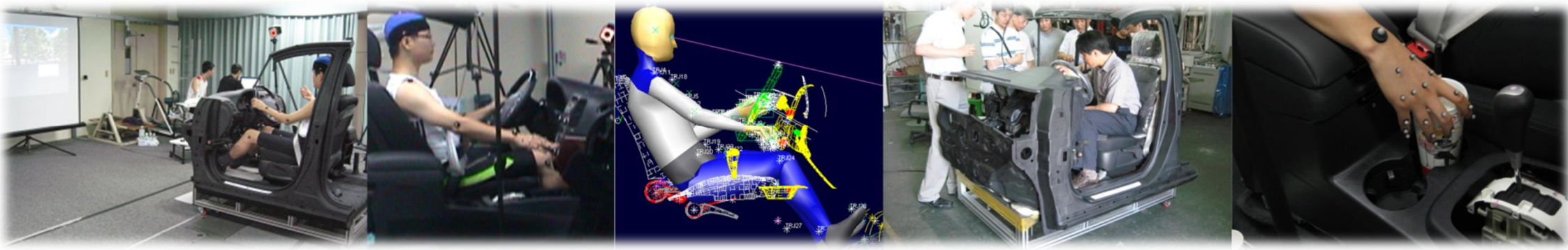
Research Methodologies
in Human Factors & Ergonomics (HF/E)



Evaluation of the digital simulation software RAMSIS for occupant packaging layout (OPL) design (You et al., 2008; sponsored by Hyundai Motor Group)

Contents

Part I

- Introduction to HF/E Research Methods
- Three Types of HF/E Research
- Field vs. Laboratory Settings
- Variables & Control of Confounds
- Participant Sampling & Power

Part II

- Human Criterion Measures
- Psychometric Requirements
- Practical Requirements of Criteria
- Statistical Techniques
- Integration & Applications
- Quiz

Learning Goals (1/2)

GOALS

- Describe the **primary purpose of HF/E research**.
- Outline the **generic process of HF/E research**.
- Distinguish among **descriptive research, experimental research, and evaluation research**.
- Compare **criterion variables with stratification variables** in descriptive research.
- Compare **between-subjects design with within-subjects design** in experimental research.

- Distinguish among **independent variables (IVs), dependent variables (DVs), and extraneous variables** in experimental research.
- Explain methods to **control confounds** in experimental research.
- Compare **formative evaluation and summative evaluation** in evaluation research.
- Evaluate the advantages and disadvantages of **laboratory settings and field settings** in HF/E research..
- Explain **the trade-off between experimental control and ecological validity** when designing an HF/E study.

Learning Goals (2/2)

GOALS

- Describe the importance of drawing a **representative sample** that properly reflects the target user population.
- Identify **biases involved in a research study** that **limit the generalizability of research findings**.
- Explain **the relationship among statistical power (1-β), effect size, type I error probability (α), and sample size** to justify research conclusions.

What is HF/E Research?

Systematic Inquiry

HF/E research is the systematic investigation into the nature of human-system interactions. It goes beyond anecdotal observation to rigorously collect and analyze data about human capabilities, limitations, and behaviors in relation to technology, tasks, and environments.

Generalizable Knowledge

The primary goal is to generate knowledge that can be generalized to broader contexts. Unlike simple usability testing of a specific product, research aims to uncover fundamental principles and guidelines that inform future designs and standards.

Multi-Methodological Approach

It encompasses a diverse range of methodologies including descriptive studies (profiling users), experimental studies (establishing causality), and evaluation research (assessing system performance), often used in complementary ways.

HF/E Research Continuum

HF/E research spans from understanding basic human capabilities to evaluating specific system designs, often in an iterative cycle.

Basic Research

Descriptive Research

"What is happening?"

- Characterize users
- Define problems
- Generate hypotheses

Applied Research

Experimental Research

"Why does it happen?"

- Test causality
- Derive principles
- Control variables

Validation

Evaluation Research

"Is it better?"

- Assess systems
- Formative (Process)
- Summative (Outcome)

Iterative Design Cycle

Understanding
Describing human limits and capabilities in context.

Evaluating
Deciding if a specific design meets requirements.

Generic HF/E Research Process

A systematic approach to ensuring validity and reliability in human-factors research.

1 Problem & Theory
- Define problem scope
- Review literature
- Formulate hypotheses

2 Measures
- Select dependent variables (DV; criteria)
- Instrumentation
- Pilot testing

3 Study Design
- Define independent variables (IV)
- Control confounds
- Develop protocol

4 Data & Analysis
- Collect data
- Screen/Clean data
- Statistical tests

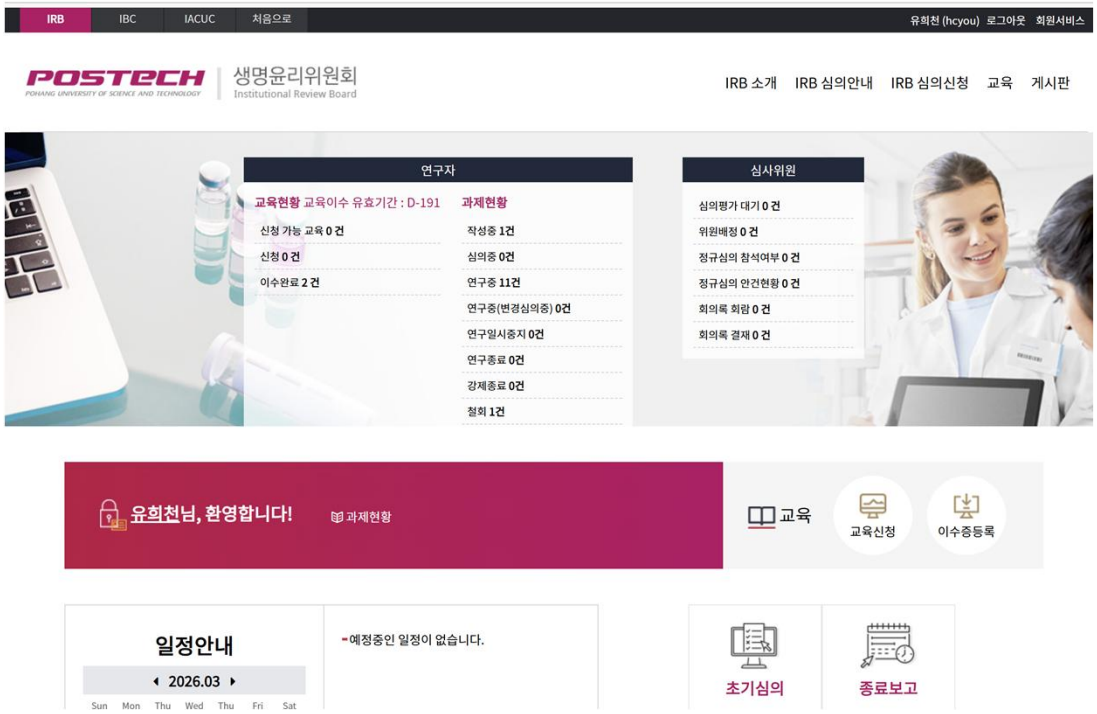
5 Reporting
- Interpret results
- Design requirements
- Disseminate

Iterative Refinement

Iterative Process | Ethical Oversight (IRB) | Stakeholder Feedback

IRB (Institutional Review Board)

An **ethics committee** that reviews and monitors research involving **human subjects to ensure their rights, safety, and welfare are protected**.



HF/E Research Types

Dimension	Descriptive Research	Experimental Research	Evaluation Research
Key Question	**"What is happening?"** Who / Where / When	**"Why does it happen?"** What causes what?	**"Is design A better?"** Does it meet criteria?
Primary Purpose	Characterize users & tasks Identify problems Generate hypotheses	Test causal relationships Establish design principles Verify theories	Assess usability / safety Compare design alternatives Acceptance testing
Control Level	**Low**: Naturalistic observation favored	**High**: Rigorous control of extraneous variables	**Moderate**: Contextual relevance is prioritized
Typical Outcome	Distributions, correlations, user profiles, task flows	Effect sizes, interaction plots, statistical significance	Pass/Fail decisions, usability scores, issues list

Research Funnel: **1. Descriptive** Scope the problem → **2. Experimental** Understand causes → **3. Evaluation** Verify solution

Descriptive Research

Research Purpose

To characterize users, tasks, environments, and behaviors as they naturally occur without manipulation. It focuses on answering "What is happening?" rather than "Why is it happening?".

When to Use

Ideal for scoping problems, defining user requirements, establishing baselines, and informing initial design requirements. Essential when the variables are unknown or cannot be controlled ethically/practically.

Key Outputs

Distributions of user characteristics (anthropometry), frequency of behaviors (error rates), correlations between variables, and rich ethnographic insights for modeling system inputs.

Descriptive Methods

Observations

Systematic recording of behavior in natural or controlled settings.

HF/E Example
Recording posture frequencies (OWAS) of assembly line workers to identify musculoskeletal risks.

Task Analysis

Decomposing activities into hierarchical goals and operations.

HF/E Example
Hierarchical Task Analysis (HTA) of a control room operator's emergency shutdown procedure.

Surveys & Interviews

Collecting self-reported data on attitudes, experiences, or demographics.

HF/E Example
Gathering user demographics and symptom prevalence of computer vision syndrome in office workers.

Archival Data Analysis

Examining existing records to identify patterns or trends.

HF/E Example
Analyzing accident reports to classify human error types in aviation incidents.

Nominal Data
Categories
(e.g., Gender, Error Types)

Ordinal Data
Rankings
(e.g., Severity Levels,)

Interval/Ratio Data
Measurements
(e.g., Time, Force)

Descriptive Variables

Criterion Variables

The outcomes being measured

Definition & Characteristics

- Primary measures of interest in the study
- Used to describe the characteristics of the population or system
- Directly reflect performance, physiology, or subjective experience

Examples

Task completion time (seconds)
Error rate (%)
Heart rate variability (HRV)
Subjective comfort rating (1-7 scale)

VS

Stratification Variables

Grouping factors for comparison

Definition & Characteristics

- Factors used to categorize data into subgroups (strata)
- Allow for comparison between different user segments
- Pre-existing characteristics (not manipulated like IVs)

Examples

Age group (Young vs. Elderly)
Expertise (Novice vs. Expert)
Gender (Male vs. Female)
Physical size (5th vs. 95th percentile)

Experimental Research

Purpose: Establish Causality

To test the causal effects of manipulated variables on human performance or experience while controlling for other factors.

- Determining "cause and effect" relationships (Does X cause Y?)
- Isolating the impact of specific design features or conditions

When to Use

Primarily used during design evaluation or theoretical testing phases:

- Comparing design alternatives (e.g., Is display A faster than display B?)
- Testing theoretical predictions about human capabilities
- Investigating interactions between variables (e.g., Noise × Complexity)

Key Outcomes

Effect Sizes
Quantifying the magnitude of difference between conditions (e.g., Cohen's *d*)

Statistical Significance
Probability that results are not due to chance (*p*-values)

Interactions
How the effect of one variable depends on the level of another

Experimental Designs: Overview

Between-Subjects Design

- Different participants in each condition (Group A vs. Group B)
- **Pros:** No learning/carryover effects; shorter session time
- **Cons:** Requires larger sample size; individual differences increase error variance
- **Example:** Testing two different keyboard layouts with two distinct groups of users

Within-Subjects Design

- Same participants experience all conditions (Condition A then B)
- **Pros:** Controls individual differences (more power); fewer participants needed
- **Cons:** Potential order/practice effects (requires counterbalancing); fatigue
- **Example:** Testing workload on three different displays with the same pilots

Factorial Designs (Interactions)

Designs with two or more independent variables (factors) to study main effects and interactions.

2×2 Design: 2 Factors,
2 Levels each (4 conditions)

Mixed Design: Combines between & within factors

Concept Visualization

A1	A2
B1	B2 (Interaction)

Experimental Variables

Independent Variable (IV)

The "Cause" / Manipulated

Key Characteristics

- ✓ Factor systematically manipulated by the researcher
- ✓ Has at least two levels (conditions)
- ✓ Assumed to influence the dependent variable

HF/E Examples

- Display Type (Digital vs. Analog)
- Lighting Level (Low vs. High)
- Device Weight (Light, Med, Heavy)

Extraneous Variable

The "Noise" / Controlled

Key Characteristics

- ✓ Factors other than IV that could affect the DV
- ✓ Must be controlled (held constant) or randomized
- ⚠ If uncontrolled, becomes a confound

HF/E Examples

- Ambient Noise (held constant)
- Time of Day (randomized)
- Subject Fatigue (counterbalanced)

Dependent Variable (DV)

The "Effect" / Measured

Key Characteristics

- ✓ Outcome measured to see the effect of the IV
- ✓ Expected to change as a result of manipulation
- ✓ Must be sensitive, reliable, and valid

HF/E Examples

- Reaction Time (ms)
- Number of Errors (count)
- Workload Rating (NASA-TLX)

HF/E Tool: NASA-TLX (Task Load Index)

NASA-TLX (Task Load Index) is a **subjective assessment tool** widely used in human factors to measure **perceived workload**. It evaluates tasks based on **six dimensions: mental demand, physical demand, temporal demand, performance, effort, and frustration**.

Rating Scale Definitions

Title	Descriptions
MENTAL DEMAND	How much mental and perceptual activity was required (e.g., thinking, deciding, calculating, remembering, looking, searching, etc.)? Was the task easy or demanding, simple or complex, exacting or forgiving?
PHYSICAL DEMAND	How much physical activity was required (e.g., pushing, pulling, turning, controlling, activating, etc.)? Was the task easy or demanding, slow or brisk, slack or strenuous, restful or laborious?
TEMPORAL DEMAND	How much time pressure did you feel due to the rate or pace at which the tasks or task elements occurred? Was the pace slow and leisurely or rapid and frantic?
PERFORMANCE	How successful do you think you were in accomplishing the goals of the task set by the experimenter (or yourself)? How satisfied were you with your performance in accomplishing these goals?
EFFORT	How hard did you have to work (mentally and physically) to accomplish your level of performance?
FRUSTRATION LEVEL	How insecure, discouraged, irritated, stressed and annoyed versus secure, gratified, content, relaxed and complacent did you feel during the task?

Place a mark at the desired point on each scale:

MENTAL DEMAND

Low — High

PHYSICAL DEMAND

Low — High

TEMPORAL DEMAND

Low — High

PERFORMANCE

Good — Poor

EFFORT

Low — High

FRUSTRATION

Low — High

Evaluation Research

Research Purpose

Systematically assess the worth, value, or merit of a system's usability, safety, and performance against defined criteria.

Key Questions

Does the system meet safety requirements?

Is Design A significantly better than Design B?

Can 95% of users complete the task without error?

Focus

Unlike experimental research which tests causal laws, evaluation focuses on specific system performance in context.

01 Formative Evaluation (Diagnostic)

Conducted **during development** to identify problems and iterate on design.

Iterative cycle: Test → Fix → Retest

02 Summative Evaluation (Validation)

Conducted **after development** to verify if benchmarks or standards are met.

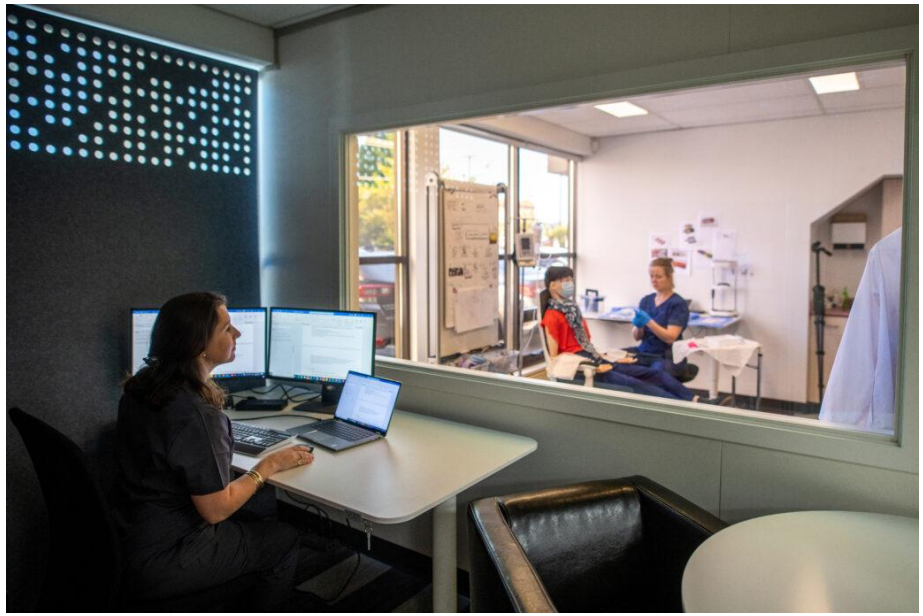
Decision: Pass / Fail

03 Primary Outcomes

Performance metrics (time, errors), usability scores (NASA-TLX, SUS), and regulatory compliance reports.

Formative Evaluation vs. Summative Evaluation

	Formative	Summative
	Help students to learn and practice	**Assess student performance**
When	Throughout the course	At the end of the instructional period
Why	Identify gaps and improve learning	Collect evidence of student knowledge, skill or proficiency
How	Via approaches that support specific student needs	Via exit learning products or a cumulative assessment

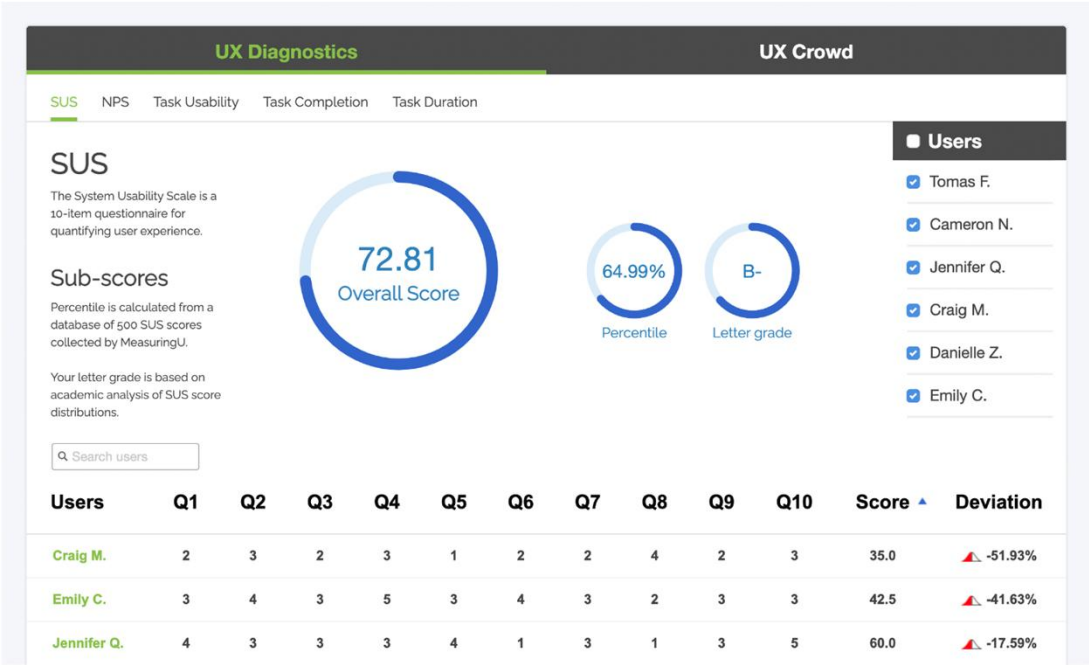


HF/E Tool: SUS (System Usability Scale)

The System Usability Scale (SUS) is a standard **10-item questionnaire** used to evaluate the **perceived usability** of products and systems. It uses a **5-point Likert scale to calculate a single score from 0 to 100**, providing a **quick, reliable measure of overall ease of use**.

System Usability Scale (SUS)

1. I think that I would like to use this system frequently.	1	2	3	4	5
2. I found the system unnecessarily complex.	1	2	3	4	5
3. I thought the system was easy to use.	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this system.	1	2	3	4	5
5. I found the various functions in this system were well-integrated.	1	2	3	4	5
6. I thought there was too much inconsistency in this system.	1	2	3	4	5
7. I would imagine that most people would learn to use this system very quickly.	1	2	3	4	5
8. I found the system very cumbersome to use.	1	2	3	4	5
9. I felt very confident using the system.	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this system.	1	2	3	4	5



Evaluation Methods & Metrics

Evaluation Methods

Usability Testing

Empirical evaluation observing representative users performing real tasks. Measures effectiveness, efficiency, and satisfaction.

Inspection Methods

Expert reviews such as Heuristic Evaluation and Cognitive Walkthroughs to identify violations of design principles.

Performance Metrics

Objective measures including Task Success Rate (%), Time-on-Task, and Error Frequency.

Key Metrics & Scales

NASA-TLX (Workload)

Multi-dimensional rating scale assessing Mental, Physical, Temporal demands, Effort, Performance, and Frustration.

System Usability Scale (SUS)

A "quick and dirty" 10-item scale yielding a single composite score (0-100) for perceived usability.

Decision Criteria

Comparing results against benchmarks (e.g., SUS > 68) or previous design iterations (Pass/Fail).

Evaluation research often combines these methods (triangulation) to provide a comprehensive assessment of system performance and user experience.

Research Setting: Field vs. Laboratory

Field Settings
Real-world environment

Key Characteristics

- ✓ **High Ecological Validity:** Findings generalize well to real-world situations.
- ✗ **Lower Control:** Hard to control extraneous variables (noise, interruptions).
- ⚠ **Complex Logistics:** Access to sites, safety, and equipment portability challenges.

Pros & Cons Summary

Captures natural behavior and context
Higher stakeholder buy-in
Risk of confounding variables

VS

Laboratory Settings
Controlled environment

Key Characteristics

- ✓ **High Internal Validity:** Strict control over variables allows causal inference.
- ✗ **Lower Realism:** Artificial setting may alter participant behavior (Demand Characteristics).
- ✓ **Easier Instrumentation:** Optimal conditions for sensitive equipment (e.g., EEG, Eye-tracker).

Pros & Cons Summary

Replicable and precise measurement
Safety for testing dangerous scenarios
Limited generalizability to complex world

Field Studies: Pros, Cons, and Use Cases

Pros

- **High Ecological Validity:** Captures natural user behavior in real environments.
- **Stakeholder Buy-in:** Results are visibly relevant to actual operations.
- **Rich Context:** Uncovers complex interactions and factors often missed in labs.

Cons

- **Low Control:** Difficult to isolate variables; susceptibility to confounds.
- **Logistical Challenges:** Often expensive, time-consuming, and hard to access.
- **Safety Risks:** Potential interference with real-world operations.

When to Use

- When **context and realism** are critical to the research question.
- For **validating** laboratory findings in the real world.
- During **exploratory phases** to identify problems or requirements.

Real-World Example

"Studying air traffic controllers in an actual tower reveals communication patterns and stress factors (e.g., background noise, team dynamics) that a sterile simulator might miss."

Naturalistic Setting

Laboratory Studies: : Pros, Cons, and Use Cases

Advantages

High Control

Precise manipulation of independent variables and isolation from environmental noise.

Repeatability

Standardized protocols ensure reproducibility by other researchers.

Sensitive Measures

Enables use of complex instrumentation (e.g., fMRI, eye-tracking) impractical in the field.

Safety

Can simulate dangerous scenarios (e.g., reactor failure) without physical risk.

Ideal Use Cases

- ✓ Testing specific theoretical predictions
- ✓ Isolating causal mechanisms
- ✓ Early-stage design concepts

Disadvantages

Artificiality

Low ecological validity; sterile environments may not reflect real-world task complexity.

Limited Generalizability

Results may not transfer perfectly to operational settings due to simplified context.

Demand Characteristics

Participants may alter behavior due to awareness of observation (Hawthorne Effect).

Key Trade-off

Laboratory studies prioritize **Internal Validity** (certainty of cause-and-effect) often at the expense of **External Validity** (applicability to real-world situations).

Updated

Hybrid & Simulated Environments

Bridging the Gap: The Best of Both Worlds?

High-Fidelity Simulators & XR

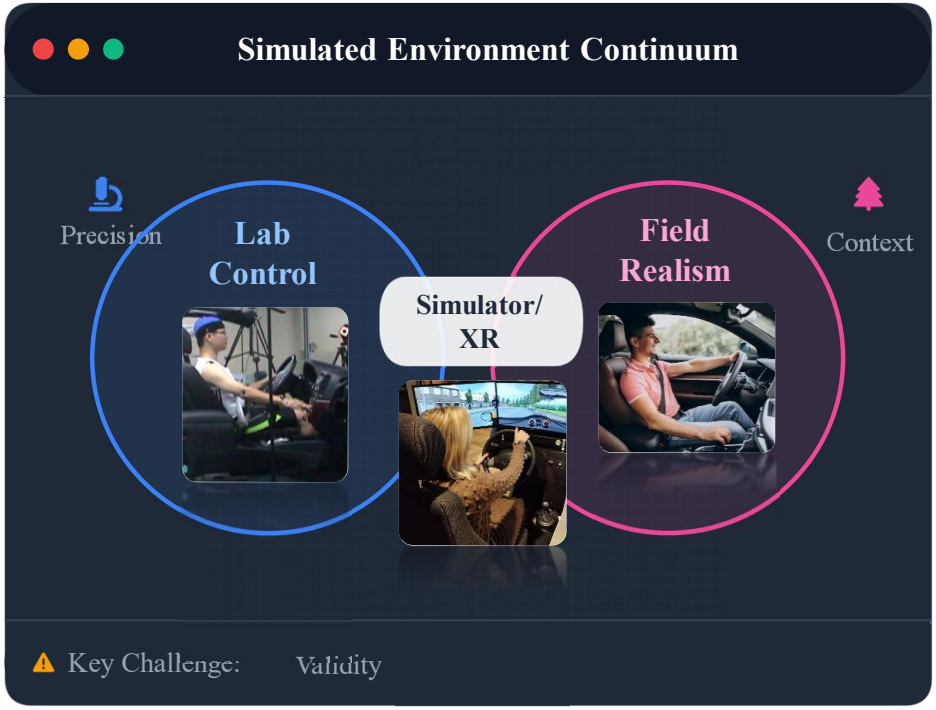
VR/AR/MR technologies allow researchers to create controlled realism, offering high ecological validity without sacrificing experimental control.

Safety in Critical Scenarios

Enables study of dangerous, rare, or critical events (e.g., pilot error, nuclear plant failure) without risking participant safety.

Measurement Capabilities

Allows for precise tracking of behavior (eye-tracking, kinematics) that would be impossible in natural field settings.



Example: Driving Simulator for Autonomous Driving Research

Asia's Best Driving Simulator in Gwangju AI Data Center (2025.09.12)

Variable Taxonomy in HF/E

01 Independent Variable (IV)

A factor manipulated or selected by the researcher to determine its effect on behavior (e.g., Display Type, Illumination Level).

02 Dependent Variable (DV)

A measure of behavior used to assess the effect of the independent variable (e.g., Reaction Time, Error Rate).

03 Criterion Variable

Used primarily in descriptive/correlation studies; the outcome being predicted or described (analogous to DV in experiments).

04 Extraneous Variable vs. Confounding Variable

Extraneous variables are nuisance factors; Confounding variables systematically covary with the IV, threatening internal validity.

Clear definition of variables is the first step in designing a valid study.

Independent Variables (IVs)

Common HF/E Examples

Systematically manipulated factors including **Display Modality** (Visual vs. Auditory), **Control Type** (Touch vs. Physical), or **Feedback Timing** (Immediate vs. Delayed).

Mechanism Mapping

IVs should target specific hypothesized mechanisms in human information processing (e.g., varying contrast to test perceptual encoding vs. varying complexity to test working memory).

Interactions & Factorial Designs

Use factorial designs (e.g., 2×2) to reveal crucial interactions, such as whether the benefit of a new display depends on the user's experience level.

Variable Classification

Equipment
Interface layout, Device type, Automation level

Environment
Lighting, Noise, Temperature, Vibration

Task
Pacing, Difficulty, Memory load

Manipulation

Independent Variables (IV)
Factors manipulated by the researcher to determine their effect on behavior or performance

Impacts DVs

Dependent Variables (DVs)

Performance Measures

Quantifiable outcomes of task execution:

Speed: Reaction time, task completion time
Accuracy: Error rates, deviation from target
Throughput: Fitts' Law index of performance (bits/s)

Physiological Indices

Objective bodily responses to strain/stress:

Cardiac: Heart Rate (HR), Heart Rate Variability (HRV)
Ocular: Blink rate, pupil diameter, scan path
Neural: EEG (alpha/theta waves), fNIRS (cortical oxygenation)

Subjective Workload

User perception of task demand:

NASA-TLX: Mental, physical, temporal demand, effort, frustration
SWAT: Subjective Workload Assessment Technique
Rating Scales: Likert scales for difficulty or comfort

Selection Criteria

Key factors for choosing valid DVs:

Sensitivity: Detects small changes in IV
Reliability: Consistent measurement over time
Validity: Actually measures the construct of interest

Insight: Multiple DVs (e.g., speed + accuracy + workload) provide a holistic view of human performance ("triangulation") and reveal potential trade-offs.

Extraneous Variable vs. Confounding Variable

Extraneous Variables

Variables other than the IV that *could* affect the DV.

- Ideally controlled or randomized across conditions
- Adds "noise" to the data but not necessarily bias
- **Example:** Participant mood, time of day (if balanced)

Confounding Variables

An extraneous variable that systematically varies *with* the IV.

- Provides an alternative explanation for the results
- **Fatal flaw** for internal validity
- Must be eliminated by design

HF/E Case Example

The "Learning Effect" Confound

Scenario: Comparing two interface designs (A vs. B) on task time.

Flawed Design (Confounded)

All participants use **Design A first**, then **Design B second**.

Design A → Design B

Result: Design B looks faster, but is it the design or practice?

Causal Confusion

IV: Interface → DV: Time

Practice Effect

Solution:
Counterbalancing (half do A then B; half do B then A).

Control Method: Randomization

Random Assignment (Between-Subjects)

The "Great Equalizer" for unknown confounds.

Assigns participants to groups such that each has an equal chance of being in any condition.

Transforms constant errors (bias) into random errors (noise), which statistics can handle.

Block Randomization

Ensures balance of sample size across conditions.

Randomize within small blocks (e.g., AABB, BABA, ABAB, BBAA).

Critical for small samples to avoid unequal group sizes by chance (e.g., AAAAA vs BBB).

Randomization of Trials (Within-Subjects)

Controls for progressive error (fatigue, learning).

Presentation order of stimuli/tasks is randomized for each participant.

Prevents systematic order effects from confounding the treatment effect.

Procedural Integrity

True randomization vs. Haphazard choice.

Use RNGs (Random Number Generators) or software (e.g., Python, R, Excel).

Seeded Randomization: Set a specific seed value to ensure the random sequence can be reproduced for verification.

Distinction: Random Selection vs. Assignment
Random Sampling (Selection) affects *External Validity* (Generalizability).
Random Assignment affects *Internal Validity* (Causality). In HF/E experiments, assignment is often more critical and controllable.

Control Method: Counterbalancing

Aspect	Complete Randomization	Partial (Restricted) Randomization	Latin Square
Concept	**All possible orders** Every sequence is administered to n participants.	**Random subset** A random selection of sequences from the total pool.	**Structured grid** Each condition appears in each ordinal position exactly once.
Sample Requirement	N = k! For 4 conditions: 24 participants For 6 conditions: 720 participants	N = Any subset Flexible sample size, independent of number of conditions.	N = Multiple of k For 4 conditions: 4, 8, 12... participants. Very efficient.
Pros & Cons	✓ Perfect control ✓ Best for few conditions (k≤4) ✗ Impossible for large k	✓ Feasible for large k ✗ Random imbalance possible ✗ Residual order effects	✓ Highly efficient ✓ Controls position effects ! Balanced square needed for sequence effects
Example (3 Conditions)	ABC, ACB, BAC BCA, CAB, CBA	Randomly select: BCA, CAB, ABC	ABC BCA CAB

Critical HF/E Nuance: Balanced Latin Square

Standard Latin Squares control for **ordinal position** (e.g., A is 1st once, 2nd once) but not **preceding items** (A is always followed by B). A *Balanced Latin Square* ensures each condition precedes and follows every other condition equally often, crucial for controlling carryover effects in fatigue or learning studies.

Balanced Square (k=4)

A	B	D	C
B	C	A	D
C	D	B	A
D	A	C	B

Control Method: Constant Conditions & Blocking

Method 1: Constant Conditions

01 Definition

Eliminating variation by holding extraneous variables at a fixed level across all treatment conditions.

02 HF/E Examples

Standardizing environmental factors (illumination, noise), equipment (display size, resolution), and procedures (scripted instructions).

03 Trade-offs

Increases internal validity (precision) but reduces external validity (generalizability). Results apply strictly to the specific fixed conditions.

Method 2: Blocking & Statistical Control

04 Blocking (Stratification)

Turning a nuisance variable into an independent variable by grouping similar units (e.g., age groups, experience levels) before random assignment.

05 Covariates (ANCOVA)

Measuring a continuous nuisance variable (e.g., baseline reaction time) to statistically remove its effect from the error variance during analysis.

Hierarchy of Control Strategies

1. **Elimination** (Constant Conditions) → 2. **Grouping** (Blocking) → 3. **Adjustment** (Covariates) → 4. **Randomization** (for all remaining unknown factors).

Protocol Checks & Fidelity

Verifying IV Manipulation

- **Definition:** Confirming that the independent variable was successfully manipulated and perceived by participants as intended.
- **Subjective Checks:** Using ratings to verify conditions (e.g., in a workload study, did the "High" condition actually elicit higher NASA-TLX scores?).
- **Physiological Checks:** Verifying arousal levels (e.g., HR increase) during stress induction tasks.
- **Construct Validity:** Essential to ensure that observed effects are due to the intended theoretical construct, not artifacts.

Fidelity & Pilot Testing

- **Task Fidelity:** The degree to which the simulated task represents the real-world domain (Physical vs. Functional vs. Psychological fidelity).
- **Pilot Testing Purpose:** A "dry run" to identify logistical issues, debug software, and check timing before full data collection.
- **De-risking:** Validates that instructions are clear and the task is not too easy (ceiling effect) or too hard (floor effect).
- **Variance Estimation:** Provides initial data to refine power analysis and sample size calculations.

Key Insight: "Type III Error"

Failing to conduct manipulation checks can lead to correctly rejecting the null hypothesis but for the wrong reasons—attributing an effect to a construct that wasn't actually manipulated as thought.

Representative Sampling for the Population

Defining the Target Population

Identify the specific user group the system is designed for. The sample must mirror this population's characteristics to ensure external validity.

Example: For a new cockpit display, recruit pilots (not students).

Sampling Frame: The accessible list or source from which the sample is drawn.

Key Stratification Variables

Ensure the sample covers relevant diversity dimensions that affect performance.

Demographics: Age, gender, anthropometry (e.g., hand size, height).

Expertise: Novice vs. expert users; domain knowledge.

Capabilities: Visual acuity, cognitive processing speed, physical strength.

Threats to Representativeness

Biases that compromise the generalizability of findings.

Selection Bias: Recruiting only volunteers or easily accessible participants (e.g., "WEIRD" samples - Western, Educated, Industrialized, Rich, Democratic).

Coverage Bias: Excluding segments of the population (e.g., elderly users in tech studies).

Random Sampling vs. Random Assignment

Random Sampling

Selection from Population

Purpose & Validity

- Ensures sample is **representative** of the larger population
- Supports **External Validity** (Generalizability)
- Rarely achieved perfectly in HF/E (often convenience samples)

HF/E Example

Selecting 100 drivers from the national driver database randomly to survey about in-vehicle display preferences.

VS

Random Assignment

Allocation to Conditions

Purpose & Validity

- Ensures groups are **equivalent** before manipulation
- Supports **Internal Validity** (Causal Inference)
- Critical for experimental control of individual differences

HF/E Example

Taking those 100 drivers and randomly assigning 50 to test Interface A and 50 to test Interface B.

Power Analysis: Concepts

Statistical Power (1-β)

Probability of correctly rejecting a false null hypothesis (detecting an effect if it exists). Aim for ≥ 0.80 (80% chance).

Type I Error Probability (α)

Probability of false positive (rejecting H_0 when true). Conventionally set at 0.05.

Effect Size

Magnitude of the phenomenon (e.g., Cohen's d). Small (0.2), Medium (0.5), Large (0.8).

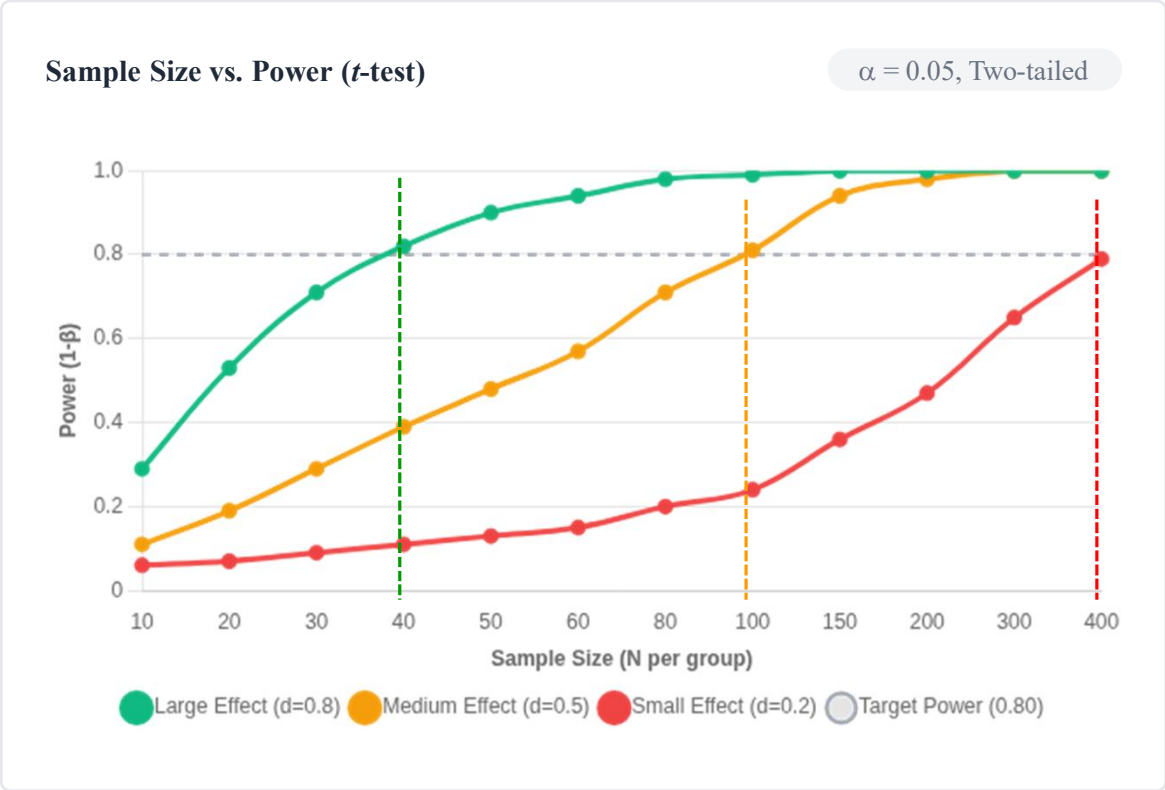
Sample Size (N)

Number of participants needed. Determined by Power, Alpha, and Effect Size.

To Increase Power ↑ = Increase Sample Size **N ↑** + Increase Effect Size **ES ↑** + Increase Alpha **α ↑**

Plan for expected attrition and missing data (typically +10-20% buffer).

Example: Power Analysis



The Trade-off

As **Effect Size** decreases, the required **Sample Size (N)** increases exponentially to maintain the same statistical power (typically 1-β = 0.80).

Cohen's *d* Benchmarks (*t*-test)

Small (*d*=0.2)	N ≈ 400 / group
Medium (*d*=0.5)	N ≈ 100 / group
Large (*d*=0.8)	N ≈ 40 / group

Recommendation

Always perform a priori power analysis using tools like G*Power to justify N before data collection.